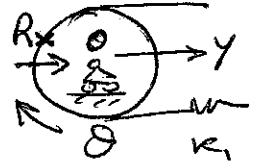


SOLUZIONE 28 gennaio 2008

con spostamento in melle

$$\underline{x} = \{x \ \varphi \ \phi \ y\}^T \quad ; \quad y = \text{spostamento orizzontale} \\ \text{cerniera O}$$

$$E_c = \frac{1}{2} M_1 \dot{y}^2 + \frac{1}{2} J_1 \dot{\theta}^2 + \frac{1}{2} M_2 \dot{x}^2 + \frac{1}{2} M_3 v_c^2 + \frac{1}{2} J_3 \dot{\phi}^2 + \frac{1}{2} M_4 v_p^2$$

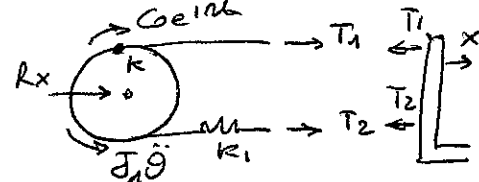


$$\dot{\theta} = \frac{\dot{x}}{R_1} - \frac{\dot{y}}{R_1} \quad ; \quad v_c = \dot{x} + R_3 \dot{\phi} \quad ; \quad v_p = \dot{x} + R_3 \dot{\phi} + L \dot{\phi}$$

$$M_F = \text{diag}([M_1 \ J_1 \ M_2 \ M_3 \ J_3 \ M_4])$$

$$[F_M] = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1/R_1 & 0 & 0 & -1/R_1 \\ 1 & 0 & 0 & 0 \\ 1 & R_3 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & R_3 & L & 0 \end{bmatrix}$$

SENZA PARTIZIONE, EQUAZIONI DI EQUILIBRIO DEL SOLO DISCO:



$$T_2 = 2k_1 x \quad - \text{FORZA EL. MOLLA } k_1 \\ \sum M_k = 0 \rightarrow G e^{i \omega t} - R_x R_1 - J_1 \dot{\theta} - T_2 2R_1 = 0 \\ \rightarrow R_x = \frac{G e^{i \omega t}}{R_1} - \left(4k_1 - \frac{\omega^2 J_1}{R_1^2} \right) x$$

$$V_{el} = \frac{1}{2} k_1 \Delta l_1^2 + \frac{1}{2} k_2 \Delta l_2^2 \quad ; \quad [k_{el}] = \text{diag}([k_1 \ k_2])$$

$$\Delta l_1 = 2x - 2y \quad ; \quad \Delta l_2 = R_3 \phi$$

$$[F_{el}] = \begin{bmatrix} 2 & 0 & -2 \\ 0 & R_3 & 0 \end{bmatrix}$$

$$V_g = M_4 g (-L \cos \phi) \rightarrow [k_g] = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & M_4 g L & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$D = \frac{1}{2} k_2 \Delta l_2^2 \rightarrow [F_D] = F_2 \quad [F_D] = [0 \ R_3 \ 0 \ 0]$$

$$\delta L = F \delta x_p + C \delta \theta = F(\delta x + R_3 \delta \phi + L \delta \phi) + C \left(\frac{\delta x}{R_1} - \frac{\delta y}{R_1} \right) + R_x \delta y$$

$$Q_{FE} = \{1 \ R_3 \ L\}^T F_0 e^{i \omega t}$$

$$Q_{CL} = \{1/R_1 \ 0 \ 0\}^T G e^{i \omega t}$$

$$Q_{cv} = -\frac{C_0 e^{i \omega t}}{R_1} + R_x$$

→ partizione delle uscite